X-SVM: An Extension of C-SVM Algorithm for Classification of High-Resolution Satellite Imagery

Giannis Lantzanakis, Zina Mitraka, and Nektarios Chrysoulakis

***Abstract*— The accurate land cover mapping of the Earth's surface using Earth observation data is one of the most studied, but yet the most challenging tasks of remote sensing field, particularly when it comes to urban areas. The large spectral variability of man-made structures, as well as the mixed pixel phenomenon, imposes the use of computational demanding techniques, which are not always effective for real case applications. Support vector machines (SVMs) are supervised learning models with associated learning algorithms, which are mainly used for classification and regression analysis. Specifically, a support vector classifier (SVC) constructs a hyperplane or a set of hyperplanes in a high-dimensional space, which separates the training data into different classes. These are then used to classify a whole image, or series of images. The current standard SVM algorithm for classification used by the most popular mapping software (e.g., ENVI, EnMAP) is the C-SVC. The parameterization of a C-SVC strongly affects the final classification result. Yet, there is no rule of thumb to choose the optimal parameters when classifying satellite imagery. Optimal parameterization totally depends on the training data, and to determine it for a specific case, a time-consuming trial-and-error process is inevitable. In this work, advancements for the C-SVC algorithm are proposed to enhance its performance when used to classify remote sensing data, eliminating the need for a part of manual parametrization, while ensuring increasing its performance.**

***Index Terms*— Land cover classification, Sentinel-2, sequential minimal optimization (SMO), support vector machines (SVMs), urban areas.**

I. Introduction

THE accurate land cover mapping of Earth's surface using Earth observation data is one of the most studied, but yet the most challenging tasks of remote sensing field, particularly when it comes to land cover mapping of urban areas where the complex form of cities imposes certain complexity [1], [2]. The large spectral variability of man-made and natural structures, as well as the mixed pixel phenomenon, imposes the use of computational demanding techniques, which are not always effective for real case applications. The remote sensing community widely uses support vector machines (SVMs) for image classification [support vector classifier (SVC)] and spectral mixture analysis [support vector regression (SVR)].

There are several examples in the literature of comparisons between SVMs and other image classification techniques applied in remote sensing data, and SVMs provide better accuracy in most cases. For example, in 2002, Huang *et al.* [3] used a spatially degraded Landsat TM image and they compared SVM with other popular classifiers, including the maximum likelihood classifier, neural network classifiers, and decision tree classifiers. They found that in most cases SVM gave more stable overall accuracies than other algorithms. In 2005, Pal and Mather [4] conducted two land cover classification experiments using multispectral (Landsat-7 ETM+) and hyperspectral (DIAS) data, respectively, for test areas in eastern England and central Spain. Their results show that the SVM achieves a higher level of classification accuracy than either the maximum likelihood classifier or the artificial neural network classifier, and that the SVM can be used for both small training data sets and high-dimensional data. In 2014, Poursanidis *et al.* [5] used Landsat OLI and Landsat TM images and they compared SVM with other popular classifiers, including the maximum likelihood classier, k-Nearest Neightbor classifier, and Feature Analyst. They found that SVM proved to be the most accurate for the purpose of urban land cover mapping. In 2018, Noi and Kappas [6] compared the performances of the Random Forest, kNN classifier, and SVM for land use/cover classification using the Sentinel-2 image data. They found that among the three classifiers and 14 subdata sets, SVM produced the highest overall accuracy (OA) with the least sensitivity to the training sample sizes, followed consecutively by RF and kNN.

SVMs are supervised learning models with associated learning algorithms, which are mainly used for classification and regression analysis. Specifically, an SVC constructs a hyperplane or a set of hyperplanes in a high-dimensional space, which separates the training data into different classes and then uses these hyperplanes to classify an image (or a series of images) accordingly. The original SVC algorithm, invented by Vapnik and Lerner [7] in 1963, was a linear classifier and if no hyperplane is found to separate completely the training data, it produces no result. In 1992, Boser *et al.* [8] suggested a way to embed nonlinearity in SVC using the so-called "Kernels" to allow more flexibility in training data separation. Kernels owe their name to the use of kernel functions, which enable the operation of SVC in a high-dimensional feature space without ever computing the coordinates of the data in that space, but rather by simply computing the inner product between all

Manuscript received March 4, 2020; revised June 19, 2020; accepted July 28, 2020. Date of publication August 28, 2020; date of current version April 22, 2021. This work was supported by the Operational Programme "Competitiveness, Entrepreneurship and Innovation" (NSRF 2014–2020) through the project " Innovative Actions in Environmental Research and Development (PErAn)" which is implemented under the Action "Reinforcement of the Research and Innovation Infrastructure," under Grant MIS 5002358 and in part by the Greece and the European Union (European Regional Development Fund). *(Corresponding author: Giannis Lantzanakis.)*

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Digital Object Identifier 10.1109/TGRS.2020.3017937

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pairs of data in the space [9]. Even with the use of Kernels, it is not always possible to separate completely the training data. The current standard SVM algorithm for classification of remote sensing data (C-SVC) used by the most popular mapping software (e.g., ENVI [10] and EnMAP [11]) was published in 1995 by Cortes and Vapnik [12]. C-SVC can perform image classification even when it fails at separating the training data completely. SVC and C-SVC are known as hard margin and soft margin classifiers, respectively. Hard margin means that the classifier has to completely separate the training data to provide a solution. On the other hand, a soft margin classifier finds the best hyperplanes that separate the training data allowing also some errors in the training data separation. This way, the soft margin classifiers always provide a result, but strongly depend on the training data and parameterization of the model.

In 1998, Platt [13] proposed an algorithm for optimization of the C-SVC, called sequential minimal optimization (SMO), to make it more efficient in terms of computational demand. Training a C-SVC requires the solution of a large quadratic programming (QP) optimization problem. The SMO breaks this large QP problem into a series of simpler possible QP problems. These small QP problems are solved analytically, which avoids using a time-consuming numerical QP optimization. Since 2000, Chang and Lin [14] have been developing the package called LIBSVM, a library for SVMs which gained wide popularity in the machine learning community. LIBSVM uses an SMO-type decomposition method and it is widely used for both image classification and regression. In 2007, Janz *et al.* [15] presented a user-oriented tool for advanced classification of hyperspectral data using the LIBSVM, known as image SVM.

The parameterization of C-SVC strongly affects the final classification result. Since there is no rule of thumb to select the optimal parameters when classifying satellite imagery, a trial-and-error process to achieve efficient separation of the training data and therefore to aim for better classification results is inevitable. To determine a good parametrization for a specific case, several tests are needed and it is even difficult to decide whether the parameterization is optimal or more attempts are necessary. Therefore, in this work, advancements for the C-SVC algorithm are proposed to enhance its performance when used to classify remote sensing data, eliminating the need for a part of manual parametrization, while maintaining or increasing its performance.

II. Theoretical Background

In remote sensing, image classification is usually performed with n-dimensional data sets, where each dimension represents an individual satellite band

$$\vec{x}^{(i)} = \left[x_1^{(i)}, x_2^{(i)}, \ldots, x_n^{(i)}\right]^T \tag{1}$$

where $\vec{x}^{(i)}$ is the ith training pixel in vector form and $x_n^{(i)}$ is the value of the nth band for the ith training pixel, which represents the surface reflectance at the nth wavelength. To better illustrate the theoretical background of this study, an example is presented here using 2-D. Fig. 1(a) graphically

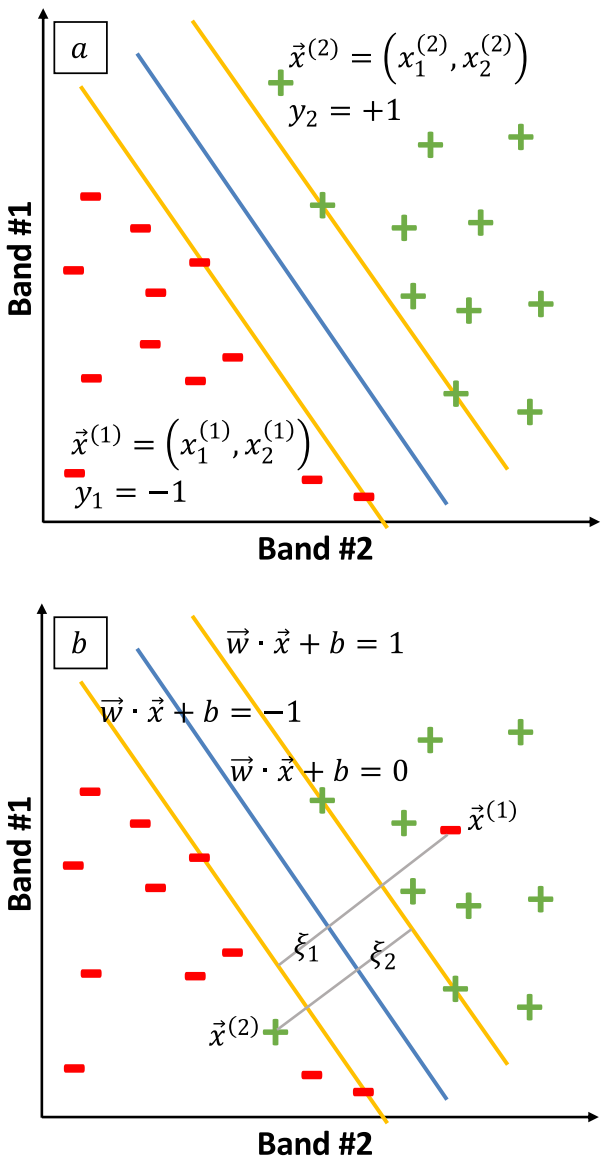


Fig. 1. 2-D example for (a) SVC and (b) C-SVC. The green crosses (+) represent the vectors that belong to one class and the red dashes (−) represent vectors of all other classes. The blue line is the optimal hyperplane which (a) separates all the green vectors from all the red vectors and at the same time has the maximum distance (ε) from both classes and (b) separates the most green vectors from the most red ones and at the same time the misclassified vectors are the shortest away (ξ_1, ξ_2) from the boundaries of their classes.

represents a 2-D example for an SVC training data separation. A training set of vectors (a subset of pixels from the image) is divided into two classes. The vectors $\vec{x}^{(i)}$ that belong to one class are shown in green color (+) and the vectors that belong to all other classes in red color (−). SVC aims at defining a hyperplane, which separates the green vectors from the red vectors. The optimal hyperplane [blue line in Fig. 1(a)] is the one that separates the green vectors from the red vectors and at the same time has the maximum distance (ε) from the boundaries of both classes [yellow lines in Fig. 1(a)].

To define the optimal hyperplane, an SVC minimizes (2) for $\vec{w}$ and b [7]

$$p = \frac{1}{2}|\vec{w}|^2 \tag{2}$$

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with constraints

$$y_i\left(\vec{w}\cdot\vec{x}^{(i)}+b\right)\geq 1 \tag{3}$$

where $y_i=+1$ for the training vectors of one class (green + in Fig. 1) and $y_i=-1$ for the training vectors of all other classes (red—in Fig. 1).

Directly solving this minimization problem is difficult because of the complexity of the constraints. The Lagrangian duality theory [16] allows simplification of this minimization problem, transforming it to a dual form

$$L=\frac{1}{2}\sum_i\sum_j a_i\cdot a_j\cdot y_i\cdot y_j\cdot\vec{x}^{(i)}\cdot\vec{x}^{(j)}-\sum_i a_i \tag{4}$$

with constraints

$$\sum_i a_i\cdot y_i=0 \tag{5}$$

$$a_i\geq 0 \tag{6}$$

where a_i are the Lagrange multipliers (LMs).

The minimization of (4) subject to (5) and (6) fails when the training vectors are not completely separated by the hyperplane, as shown in Fig. 1(b). In this case, (4) is decreasing indefinitely and never stops at a solution. To overcome this limitation, Cortes and Vapnik [12] presented the soft margin classifier C-SVC, which can work with the training data that are not completely separated by a hyperplane. The difference with (2) is that the C-SVC introduces an "error-distance" ξ_i for each training vector $\vec{x}^{(i)}$ in (2), which leads to the following equation:

$$p=\frac{1}{2}|\vec{w}|^2+C\cdot\sum_i\xi_i \tag{7}$$

with constraints

$$y_i\left(\vec{w}\cdot\vec{x}^{(i)}+b\right)\geq 1-\xi_i \tag{8}$$

$$\xi_i\geq 0 \tag{9}$$

where C is a given constant.

Applying again the Lagrangian duality theory transforms the problem of the dual form to the following equation:

$$L=\frac{1}{2}\sum_i\sum_j a_i\cdot a_j\cdot y_i\cdot y_j\cdot\vec{x}^{(i)}\cdot\vec{x}^{(j)}-\sum_i a_i \tag{10}$$

with constraints

$$\sum_i a_i\cdot y_i=0 \tag{11}$$

$$0\leq a_i\leq C. \tag{12}$$

The difference between SVC and C-SVC in dual form is the upper limit of the LMs. For a C-SVC, the upper limit is the value of the parameter C, which needs to be determined a priori, in contrast to the SVC where the upper limit for the LMs does not exist. Worth mentioning that for $C\to\infty$, the C-SVC tends to SVC. There is no rule of thumb to choose the optimal value for C when classifying satellite imagery. The optimal value of C totally depends on the training data set, and to determine it for a specific case, a time-consuming trial-and-error process is inevitable. For a given value of C, the optimal LMs a_i can be identified by minimizing (10) subject to (11) and (12), using the SMO algorithm [13]. If $a_i\neq 0$, the respective vector $\vec{x}^{(i)}$ from the training data set will affect the resulting hyperplane and it is called a support vector. If $a_i=0$, the respective vector $\vec{x}^{(i)}$ is actually excluded from computations, which means that it will not affect the resulting hyperplane. The optimal hyperplane that was defined using the training data set will be used to classify all the pixels in the image using the following equation:

$$\vec{w}\cdot\vec{x}+b\geq 0 \tag{13}$$

where

$$\vec{w}=\sum_{k=1}^{N}a_k\cdot y_k\cdot\vec{x}^{(k)} \tag{14}$$

$$b=\frac{1}{N}\sum_{k=1}^{N}\left(y_k-\vec{w}\cdot\vec{x}^{(k)}\right) \tag{15}$$

where $\vec{x}^{(k)}$ are the N vectors remained from minimization with SMO, that is, the support vectors (SVs).

The optimal hyperplane separates the data set into two classes. In the case of satellite imagery, we usually need to discriminate between more than two classes. For n number of classes, the separation is performed repetitively, n times. Every class is separated from all other classes and the approach is known as one–vs–all (OVA) [17], resulting in n equations

$$D^{(m)}(\vec{x})=\vec{w}^{(m)}\cdot\vec{x}+b^{(m)} \tag{16}$$

where

$$\vec{w}^{(m)}=\sum_{k=1}^{N}a_k^{(m)}\cdot y_k^{(m)}\cdot\vec{x}^{(m,k)} \tag{17}$$

$$b^{(m)}=\frac{1}{N}\sum_{k=1}^{N}\left(y_k^{(m)}-\vec{w}^{(m)}\cdot\vec{x}^{(m,k)}\right) \tag{18}$$

where $\vec{x}^{(m,k)}$ are only the final SVs for class m, where $m=1,2,\ldots,n$. Equation (16) is then applied for every single pixel $\vec{x}$ of the image and it is assigned to class m with the maximum value of $D^{(m)}(\vec{x})$.

The linear classifier described above is not suitable for most cases, since usually the training data sets are not well-separated by linear features (hyperplanes). As explained earlier in Section I, nonlinearity is embedded in SVMs using kernel functions [8], [9]. The use of kernels in C-SVC transfers the training vectors to a new vector space in a way to be separated there by a hyperplane. The most commonly used kernel functions in [14] and commercial software are as follows.

1) Linear kernel [8]

$$K\left(\vec{x}^{(i)},\vec{x}^{(j)}\right)=\vec{x}^{(i)}\cdot\vec{x}^{(j)}. \tag{19}$$

2) Polynomial kernel [18]

$$K\left(\vec{x}^{(i)},\vec{x}^{(j)}\right)=\left(\vec{x}^{(i)}\cdot\vec{x}^{(j)}+c\right)^n \tag{20}$$

where c and n are given constants.

3) RBF kernel [19]

$$K\left(\vec{x}^{(i)},\vec{x}^{(j)}\right)=\exp\left(-g||\vec{x}^{(i)}-\vec{x}^{(j)}||^2\right) \tag{21}$$

where g is a given constant.

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The kernel functions transform (10) to the following equation:

$$L = \frac{1}{2}\sum_i \sum_j a_i \cdot a_j \cdot y_i \cdot y_j \cdot K\left(\vec{x}^{(i)}, \vec{x}^{(j)}\right) - \sum_i a_i \quad (22)$$

and each pixel of the image will be placed to the mth class which maximizes the following equation:

$$D^m(\vec{x}) = \vec{w}^{(m)} \cdot \vec{\Phi}^{(m)}(\vec{x}) + b^{(m)} \quad (23)$$

where

$$\vec{w}^{(m)} = \sum_{k=1}^{N} a_k^{(m)} \cdot y_k^{(m)} \cdot \vec{\Phi}^{(m)}\left(\vec{x}^{(m,k)}\right) \quad (24)$$

$$b^{(m)} = \frac{1}{N}\sum_{k=1}^{N}\left(y_k^{(m)} - \vec{w}^{(m)} \cdot \vec{\Phi}^{(m)}\left(\vec{x}^{(m,k)}\right)\right) \quad (25)$$

where $\vec{x}^{(m,k)}$ are only the SVs for the mth class. Worth mentioning that the vector function $\vec{\Phi}^m$ can be different for each different class. In most cases, like in the cases of RBF and polynomial kernels, the vector function $\vec{\Phi}^m$ cannot be explicitly identified. By substituting (24) and (25) in (23), (23) becomes

$$D^m(\vec{x}) = \sum_{k=1}^{N}\left(a_k^{(m)} \cdot y_k^{(m)} \cdot K^{(m)}\left(\vec{x}, \vec{x}^{(m,k)}\right)\right) + \frac{1}{N}\sum_{k=1}^{N}\left(y_k^{(m)} - \sum_{K=1}^{n} a_k^{(m)} \cdot y_k^{(m)} \cdot K^{(m)}\left(\vec{x}^{(m,k)}, \vec{x}^{(m,k)}\right)\right) \quad (26)$$

which makes (26) dependent only on the inner product $\vec{\Phi}_i, \vec{\Phi}_j$ which is known from (19) to (21) and independent of $\vec{\Phi}$ which can be unspecified. Therefore, each pixel of the image will be placed to the mth class which maximizes (26).

During minimization of (22) using SMO, the LMs a_i are estimated two by two in an inner loop (IL) for all the possible pairs. This is possible because (22) is the sum of inner products of vectors $\vec{\Phi}^{(i)}(\vec{x})$ and $\vec{\Phi}^{(j)}(\vec{x})$, and therefore the problem is quadratic [13]. Specifically, solving, for example, for two LMs a_i and a_j, SMO results in

$$a_j = a_j^* + y_j \cdot \frac{(E_i - E_j)}{\eta} \quad (27)$$

$$a_i = a_i^* - y_i \cdot y_j \cdot \left(a_j - a_j^*\right) \quad (28)$$

where a_i^* and a_j^* are the two LMs from the previous iteration (usually their initial value is set to 0 for the first iteration), and y_i and y_j are set to $+1$ or -1 depending on the class that the training vector $\vec{x}^{(i)}$ belongs to. In addition

$$\eta = \frac{\partial^2 L}{\partial a_j^2} \quad (29)$$

$$E_i = \sum_{\lambda=1}^{M} a_\lambda \cdot y_\lambda \cdot \vec{x}^{(\lambda)} \cdot \vec{x}^{(i)} - y_i \quad (30)$$

$$E_j = \sum_{\lambda=1}^{M} a_\lambda \cdot y_\lambda \cdot \vec{x}^{(\lambda)} \cdot \vec{x}^{(j)} - y_j \quad (31)$$

where L is the Lagrange from (22) and M is the total number of all training vectors. Usually, η is a positive number. In unusual cases, where η is negative or 0, (27) and (28) are not applicable. In these cases, a_j is estimated by

$$a_j^{(\min)} = = \begin{cases} \max\left(0, a_j - a_i\right), & y_j \cdot y_i = -1 \\ \max\left(0, a_i + a_j - C\right), & y_j \cdot y_i = 1 \end{cases} \quad (32)$$

$$a_j^{(\max)} = \begin{cases} \min\left(C, C + a_j - a_i\right), & y_j \cdot y_i = -1 \\ \min\left(C, a_i + a_j\right), & y_j \cdot y_i = 1 \end{cases} \quad (33)$$

and a_i is calculated from (28) as a function of a_j [13].

An external loop (EL) examines the remaining pairs of vectors toward minimization of (22).

III. Methodology and Data

This section describes the extension of the C-SVC, for introducing a more effective separation of the training data, which increases or maintains the accuracy of the image classification without the need to determine the value of the parameter C. The modifications do not involve any changes in the algorithm of C-SVC, but they are based on an advantage that existed in SMO. Because of the way SMO is structured, the intermediate steps of the algorithm implementation may provide more accurate results than those of the final solution which minimizes (22). This is probably due to the fact that as long as the value of the parameter C is not optimal, minimization will not lead to the optimum possible solution. Our method finds these intermediate results and replaces the previous ostensibly optimal solution with the best possible one. We call the new classifier X-SVC.

A. SVM Independent of the Parameter C

As explained in Section II, C-SVC accepts an "error-distance" ξ_i for each training vector $\vec{x}^{(i)}$ and minimizes (7) subject to constraints (8) and (9). In the final dual form (22) ξ_i were eliminated and their impact is reflected as an upper limit of the LMs with the parameter C. The optimal solution is the one that minimizes (22) subject to constraints (11) and (12). The selection of the optimal value of the parameter C (a) is dependent on the training data and (b) impacts the classification result. The user must define a value of C to run a C-SVC. Nevertheless, so far, no robust way exists to select the optimal value C for each case and not even some rule of thumb. X-SVC keeps the LMs estimated from each EL of the SMO and decides on the optimum solution based on the separation efficiency (SE).

We define the SE as the fraction of the training vectors classified successfully

$$\mathrm{SE} = \frac{f_p + f_q}{2} \quad (34)$$

where f_p and f_q are the fractions of correctly classified training vectors corresponding to classes p and q, respectively. The optimum solution is identified based on the higher SE, which may be found between iterations while minimizing (22) and not on the final iteration, as assumed so far in C-SVC. The need for introducing a metric for comparing the percentage of

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correctly classified training vectors (instead of comparing the total correctly and incorrectly classified training vectors) arises from the OVA method. With OVA, each class is separated by all other classes, which are assumed as another single class. Therefore, OVA imposes an imbalance in the number of training vectors per class included in the separation process. If, for example, we have four classes in total in a classification scheme (i.e., A, B, C, and D), then with OVA, class A is separated from $B + C + D$, then B from $A + B + C$, then C from $A + B + D$, and finally D from $A + B + D$. If originally each class contains the same number of training vectors, then in a separation process, one class would contain three times more vectors than the other class. Therefore, it is important to account for a metric which normalizes for the number of vectors and does not account only their absolute number.

Fig. 2 shows the pseudocode for the implementation of SMO in X-SVM. Fig. 2 highlights the difference between C-SVM and X-SVM in the attempt to minimize (22).

Fig. 3(c) and (d) shows the SE (blue line) and L from (22) (red line) in the process of minimizations in relation to the number of the ELs of SMO, defining the value of C at 200 000, for two 3-D randomly generated training data sets [Fig. 3(a) and (b)]. The black dots correspond to the solution of X-SVM and C-SVM, which are the optimum solution in terms of SE for X-SVC and the minimum of (22) for C-SVC. Fig. 3(e) and (f) shows the SE for X-SVC (red line) and C-SVC (blue line) in relation to the value of C parameter in a range from 0 to 300 000. X-SVC always has the same or higher SE than C-SVC for any possible value of C. Most importantly, the SE of X-SVC seems to stabilize to the highest possible value, while C-SVC is ranging with the parameter C. The minimum value of C that tends to stabilize the SE varies with different data. To overcome searching for an optimum C by trial and error when using C-SVC, a large value of C could be used (e.g., 300 000). This ultimately leads to a fixed and maximum value of SE, without having to search for the optimal value parameter C, as opposed to C-SVC where the final result is affected from the value of C. Keeping the solution with the highest SE and not the one that minimizes (22) increases the accuracy of image classification (as demonstrated with real data in Section III-B).

In [20], it is often mentioned that a high value of C can lead C-SVC to an overfitted result. If the overfitting is only caused by the C value, X-SVC overfitted solutions are not accounted for, since X-SVC searches for the solution that maximizes the SE (which typically contains lower values than C-SVC) and not the most fitted one. Also, for a possible value of C and after, X-SVC is not affected any further, in contrast to C-SVC where the LMs have the same order of magnitude as the C parameter [Fig. 3(e)–(f)]. However, if the overfitting is caused by the selection of the kernel function, X-SVC will act the same with C-SVC as far as the overfitting is concerned.

```
while LMs change values
   for i = 1 to Total Number of SVs
      for j = 1 to Total Number of SVs
         if ( i == j )
            break;
         end
         (Note: Here are the Inner Loops)
          implementation of SMO equations
      end
   end
   (Note: Here are the External Loops)
   Compute b
   Compute Loss Function and SE
   Save LMs, SVs, b, y, Loss Function, SE
end
Keep the LMs, SVs, b and y with the highest SE
```

Fig. 2. Pseudocode for the implementation of SMO in X-SVC. The bold lines represent the lines that are added from C-SVC to X-SVC minimization. For X-SVC, in every external loop the SE is calculated using the current SVs, LMs, b and y, and the final solution is the one with the highest SE. For a C-SVC, the bold lines do not exist and the final solution is the one in the last iteration, which minimizes the loss function (22).

B. Satellite Data

A Sentinel-2 Level-2A [21] atmospherically corrected image is used in this study, acquired on a clear sky day over Heraklion, Greece (03.05.2019, 9:30 UTC). The scene covers an extent of 13 × 7 km^2 over the broader area of the city, shown in Fig. 4, and it includes a wide range of surface cover, that is, buildings, roads, industrial areas, parks, an airport, and agricultural surfaces, mainly olive trees and vineyards. The Sentinel-2 Level-2A image contains surface reflectance information covering 12 spectral bands from 442 to 2200 nm at 10 m (red, green, blue, NIR), 20 m (4 × red edge, 2 × SWIR), and 60 m (coastal aerosol, water vapor) spatial resolution. In this study, all the 20- and 60-m spectral bands were downscaled to 10-m spatial resolution using the super resolution for the Sentinel-2 images [22], to allow experiments with combination of different bands in the same spatial resolution. Reference land cover data are used for accuracy assessment. The reference data are collected from open street map (OSM) [23] data corresponding to water surfaces, buildings, and paved surfaces, with photointerpretation of the corresponding areas in the Sentinel-2 image. Vegetated areas and surfaces covered by bare soil were identified by visual inspection of the Sentinel-2 image directly and ancillary higher resolution Planet imagery acquired on 02.05.2019, 8:44 UTC. Fig. 5 shows the location of the reference data in the study area.

C. Experimental Design

An experiment was designed to assess the performance of X-SVC compared with C-SVC in image classification using single image pixels for training. Five land cover classes are assumed, that is, buildings, paved surfaces, vegetation, bare soil, and water. The classification scheme was selected based on the need arising of urban studies, particularly for urban climate [2] A number of 1000 Spectra corresponding to those classes (200 for each class) are collected from the Sentinel-2 image with visual inspection of the image and higher resolution imagery (Planet acquired on May 2, 2019)

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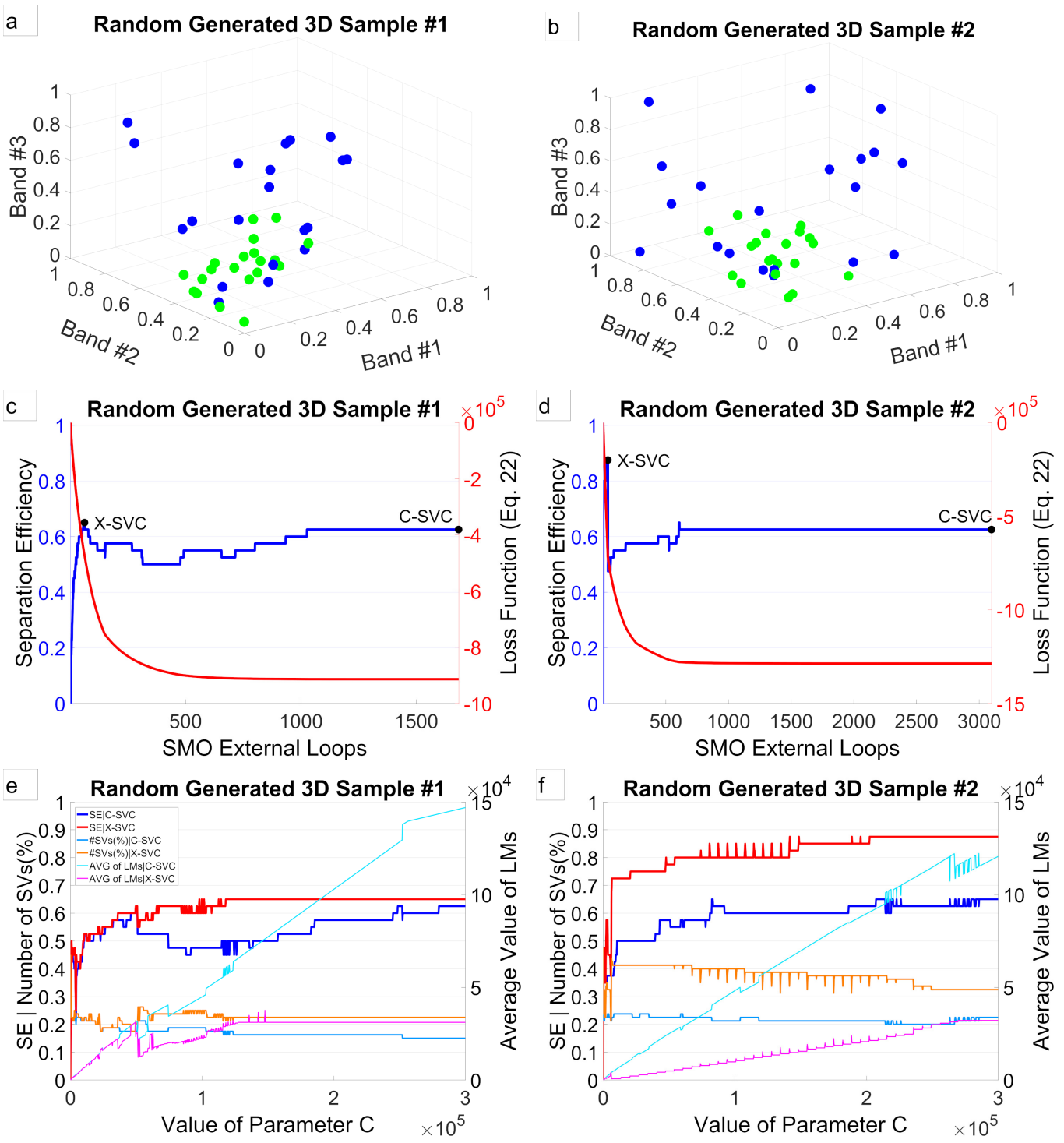


Fig. 3. Two randomly generated 3-D samples were examined to demonstrate the performance of X-SVM compared with C-SVM. (a) and (b) Randomly generated points in the 3-D vector space, for sample #1 and sample #2, respectively. (c) and (d) How the SE (blue line) and the loss function (red line) change during SMO optimization, for sample #1 and sample #2, respectively. For C-SVC, the solution is the one at the end of minimization (indicated with a black dot), while for X-SVC the solution is the one with the maximum SE (indicated with a black dot), which can be found in intermediate loops. (e) and (f) Relationship between SE, the percentage of remaining SVs, and the average value of the remaining LMs in relationship with the value of the parameter C, for both C-SVC (blue, deep sky-blue, and electric blue lines, respectively) and X-SVC (red, orange, and ping lines respectively) for sample #1 and sample #2, respectively. X-SVC always has the same or higher SE than C-SVC for every different value of the parameter C, but the most important thing is that for a high value of C and after the SE is kept constant.

as ancillary source. It has to be noted here that the training data selection process is totally independent from the reference data one. The training data collected represents one of the five classes, even in the case of mixed pixels. Spectra from pixels corresponding to more than one land cover classes in higher spatial resolution imagery (Planet) are used as training data, being assigned to the class with the larger areal coverage in the pixel.

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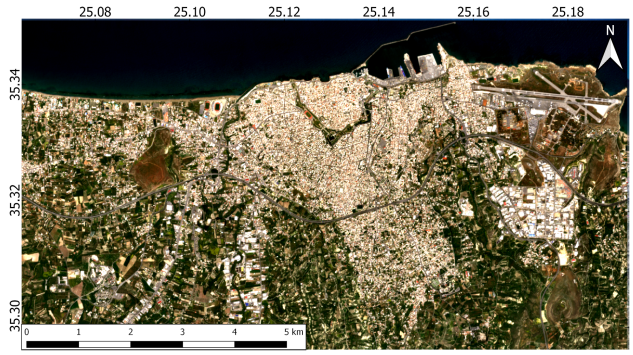


Fig. 4. True-color composition (4-3-2) of the Sentinel-2 Level-2A image of the study area, the broader area of Heraklion, Greece (May 3, 2019, 9:30 UTC).

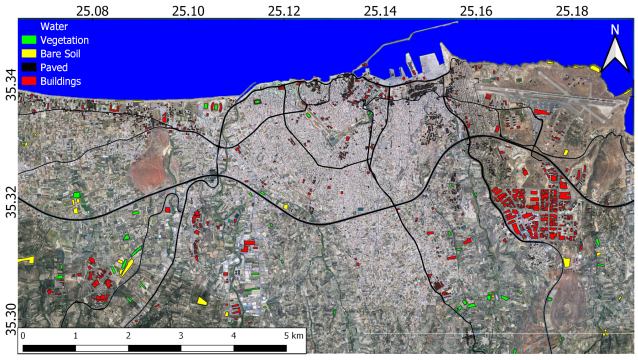


Fig. 5. Google Earth image of the study area, the broader area of Heraklion, Greece. The reference data represented with red color for buildings, blue for water, green for vegetation, yellow for bare soil, and black for paved.

The objective of the experiment is to compare the performance of X-SVC and C-SVC in image classification under the same conditions using Sentinel-2 imagery. We have to note here that the best possible classification result is affected from many factors, for example, the quantity of the training data set and the atmospheric correction, which are not considered in this work, because the emphasis here is on the comparison between X-SVC and C-SVC. To assess and compare the performance of X-SVC and C-SVC under different conditions, eight different combinations of the Sentinel-2 bands are used, for three different types of Kernels (linear, polynomial, and RBF) shown in Table I. For a two-band approach (2B), #4 (red, 10 m) and #8 (NIR, 10 m) are used, because these correspond to the ones used for NDVI, and therefore they succeed in discriminating healthy vegetated from other surfaces. For the three-band approach (3B), band #3 (green, 10 m) is added, and then, for the four-band approach (4B), band #2 (blue, 10 m) is added. To this end, for tests 2B, 3B, and 4B, the pixel information corresponds to the satellite measurement representing the 10-m area before the downscaling. The 20- and 60-m spatial resolution bands were downscaled to 10-m spatial resolution using the super resolution for the Sentinel-2 images [22] to match the 10-m resolution bands. For the six- (6B), eight- (8B), and ten (10B)-band approaches, band #11 and #12 (SWIR, 20 m), #5 and #6 (red edge, 20 m), and #7 and #8a (red edge, 20 m) are included, respectively. Therefore, for tests 6B, 8B, and 10B, the pixel information may correspond to the measurement, in case of homogeneous

TABLE I

COMBINATION OF BANDS FOR THE EIGHT EXPERIMENTS DESIGNED TO ASSESS THE PERFORMANCE OF C-SVM AND X-SVM WITH DATA IN DIFFERENT DIMENSIONS

Experiment	2B	3B	4B	6B	8B	10B	11B	12B
Band #1 (443 nm) - Coastal Aerosol (60m)							√	√
Band #2 (490 nm) - Blue (10m)			√	√	√	√	√	√
Band #3 (560 nm) - Green (10m)		√	√	√	√	√	√	√
Band #4 (665 nm) - Red (10m)	√	√	√	√	√	√	√	√
Band #5 (705 nm) - Vegetation Red Edge (20m)					√	√	√	√
Band #6 (740 nm) - Vegetation Red Edge (20m)					√	√	√	√
Band #7 (783 nm) - Vegetation Red Edge (20m)						√	√	√
Band #8 (842 nm) - NIR (10m)	√	√	√	√	√	√	√	√
Band #8A (865 nm) - Vegetation Red Edge (20m)						√	√	√
Band #9 (945 nm) - Water Vapour (60m)								√
Band #11 (1610 nm) - SWIR (20m)				√	√	√	√	√
Band #12 (2190 nm) - SWIR (20m)				√	√	√	√	√

surfaces larger than 20 m × 20 m, but some error may be introduced for nonhomogeneous areas. For tests with 11 (11B) and 12 (12B) bands, band #1 (coastal aerosol, 60 m) and #9 (water vapor, 60 m) are included, respectively, corresponding to the atmospheric bands of 60-m spatial resolution.

The spatial downscale is expected to introduce some error in the respective experiments. It is intentionally selected to downscale everything in the finest spatial resolution, for testing the performance of X-SVC in classifying Setinel-2 imagery corresponding to an urban site. For the selected classification scheme, it is important to achieve the highest spatial resolution possible, to capture successfully features such as buildings and smaller roads. The performance of C-SVC and X-SVC was assessed in terms of SE, as well as the accuracy of the derived land cover maps. The reference data described in Section III-B. were used to asses the accuracy of the land cover maps using a random Sample of 1000 pixels for each class.

IV. RESULTS AND DISCUSSION

The training data set described in Section III-C. was used to perform land cover classification for the study area (Fig. 4) with both C-SVC and X-SVC, using different band combinations and kernel functions as explained in Section III. The SE and classification accuracy were estimated in all cases and the results are presented below.

Fig. 6 shows the SE of the application of C-SVC (dotted lines) and X-SVC (solid lines) in the training data for (a) linear, (b) polynomial, and (c) RBF kernels, for the eight different combinations of the Sentinel-2 bands (Table I), for the five classes. For the application of C-SVC, ten values of the parameter C were used for testing (0.0001, 100, 1000, 5000, 10 000, 100 000, 150 000, 200 000, 250 000, and 300 000) and the one resulting in higher SE is shown in Fig. 6. It has to be noted here that C-SVC had to be executed ten distinct times to produce different results from which only the best (overall higher SE) is presented in Fig. 6, while X-SVC is executed only once (since the parameter C is set to a high value, e.g., 300 000).

Overall, X-SVC presents the same or higher SE than C-SVC in all cases. The SE increases with the inclusion of more bands for both C-SVC and X-SVC using RBF or polynomial kernel for all classes, which is not the case with linear kernel.

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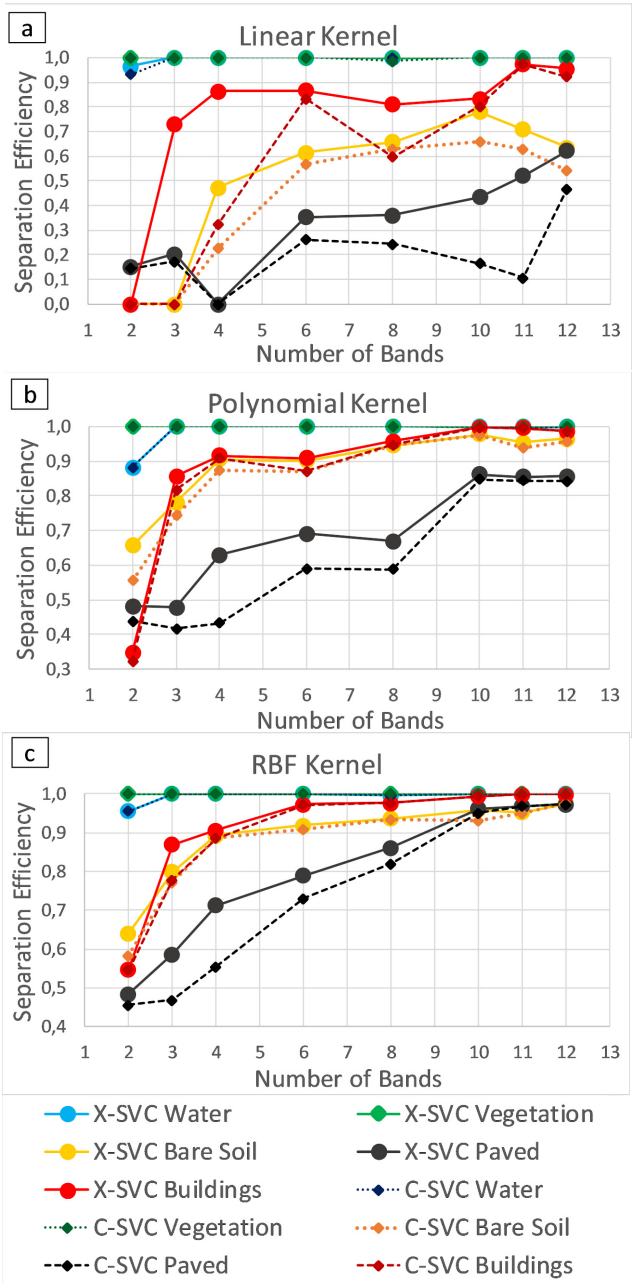


Fig. 6. SE of C-SVC (dotted lines) and X-SVC (solid lines) using (a) linear, (b) polynomial, and (c) RBF kernels for the eight different combinations of the Sentinel-2 bands (Table I), for the five classes (water, vegetation, bare soil, paved, and buildings).

Increasing the data dimensionality provides more information to the models and in this case allows better separation of the training data and increased the SE. Exceptions occur with the inclusion of band 1 and band 9 in experiments 11B and 12B. This is because of the information captured in those bands, which is atmospheric, and therefore it is not expected to be useful for the discrimination of land cover features. Moreover, the information from bands 1 and 9 originally corresponds to 60-m spatial resolution, and the downscale performed in this study assigns an estimated value to the 10-m pixel corresponding to the broader area. Therefore, tests 11B and 12B are not expected to allow further conclusions.

The SE always increases for the linear kernel for X-SVC, while for C-SVC the SE presents fluctuations. The reason is that it is very unusual for real data to be linearly separable. Therefore, a linear kernel more easily fails in separating the training data compared with the other kernels. This is a good example for demonstrating the advancement of X-SVC presenting the potential of X-SVC in separating the training data. The reason X-SVC performs better with the linear kernel in terms of SE is that it keeps the optimum solution, which is not dependent on the selection of the parameter C. The same happens with every kernel with X-SVC, and therefore X-SVC performs better or equally good with C-SVC, in case the optimum C is manually selected.

Furthermore, Fig. 6 demonstrates the great potential of X-SVC for urban land cover mapping. Surfaces corresponding to water and vegetation are already separated well with information from a few bands (SE $>$ 0.98 for 3B), both using C-SVC and X-SVC. However, X-SVC presents a remarkable performance in terms of SE compared with C-SVC for the paved and building classes, for all three Kernels, reaching an improvement of up to 0.7 (SE) for the linear kernel. This is particularly useful for urban mapping applications, considering the difficulties in performing urban land cover classification using satellite data. It is rather difficult to discriminate between the different manmade materials and it is very challenging to separate buildings from paved surfaces with the use of Sentinel-2-like satellite data, both because of the low spatial resolution relatively to the underlying features and the low spectral resolution relatively to the spectral similarity of manmade materials.

The successful separation of the training data, that is, the high SE, is a prerequisite for accurate land cover classification, but it cannot guarantee a high classification accuracy. To assess the performance of X-SVC and C-SVC in terms of classification accuracy, the SVs arising for all the experiments of Table I were used to classify the Sentinel-2 image (Fig. 4) and the classification accuracy was estimated using the individual reference data (Fig. 5) described in Section III-B.

Fig. 7 shows the OA of the land cover maps produced with C-SVC and X-SVC with the (a) linear, (b) polynomial, and (c) RBF kernels for the eight different combinations of the Sentinel-2 bands (Table I). X-SVC presents higher or equal OA than C-SVC in all cases. The polynomial and RBF kernels perform better than the linear one, as expected, following the behavior of the respective SE. The highest OA for X-SVC is 0.867 (RBF for 8B) and the highest OA for C-SVC is 0.848 (RBF for 8B). The highest difference in OA between X-SVC and C-SVC is 0.293 (RBF for 2B). The lowest OA for X-SVC is 0.563 (polynomial for 2B) and for C-SVC is 0.462 (polynomial for 2B).

It has to be noted here that the OA does not increase with the inclusion of information from more bands, as opposed to the SE. The highest OA is found for experiments 6B, 8B, and 10B, as opposed to the SE. Increasing the training data dimensionality may improve in some cases the SE as shown earlier. However, this has no positive impact on classification accuracy. The information included from bands 1 and 9 is not appropriate for land cover classification, as stated earlier.

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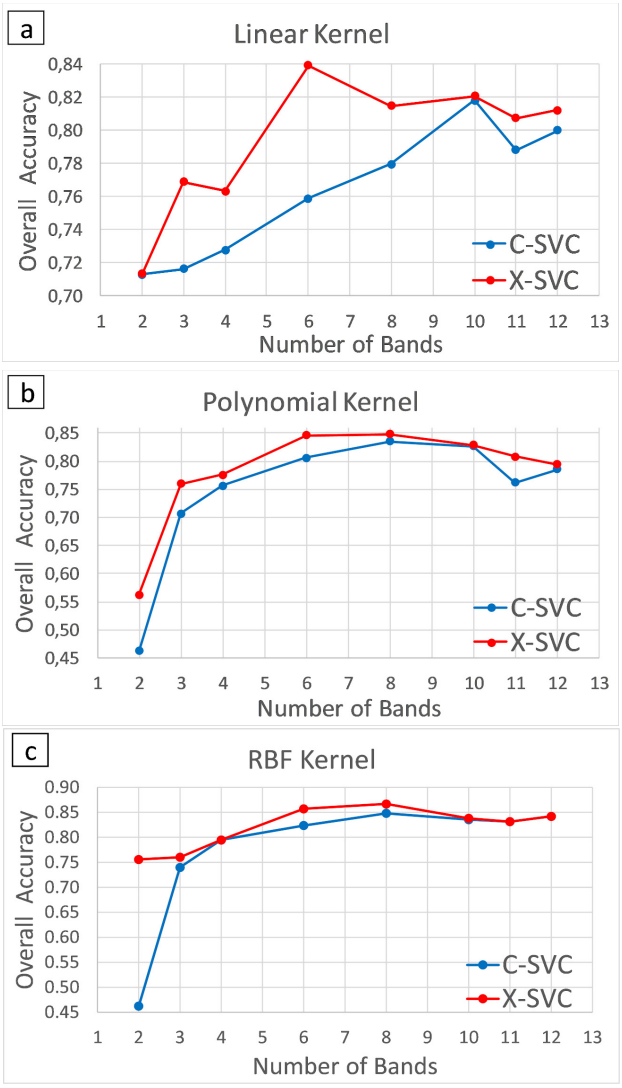


Fig. 7. OA for the (a) linear, (b) polynomial, and (c) RBF kernel for eight different combinations of the Sentinel-2 bands (Table I), for C-SVC (blue lines) and X-SVC (red lines).

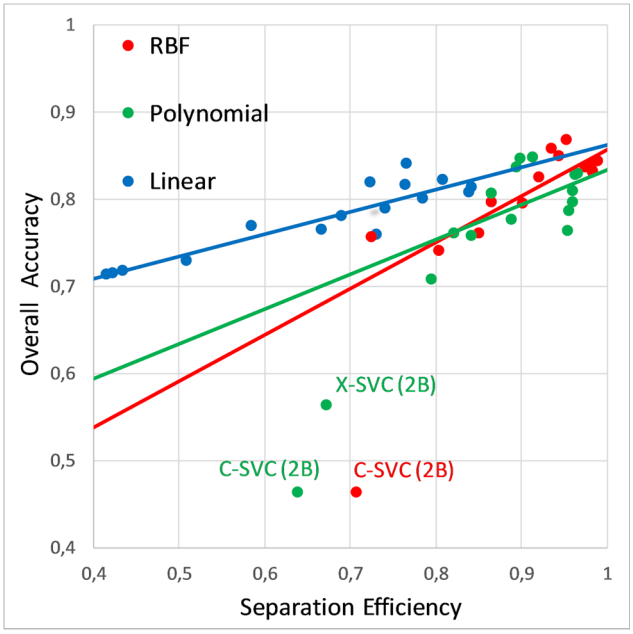


Fig. 8. SE versus OA for all the experiments of Table I.

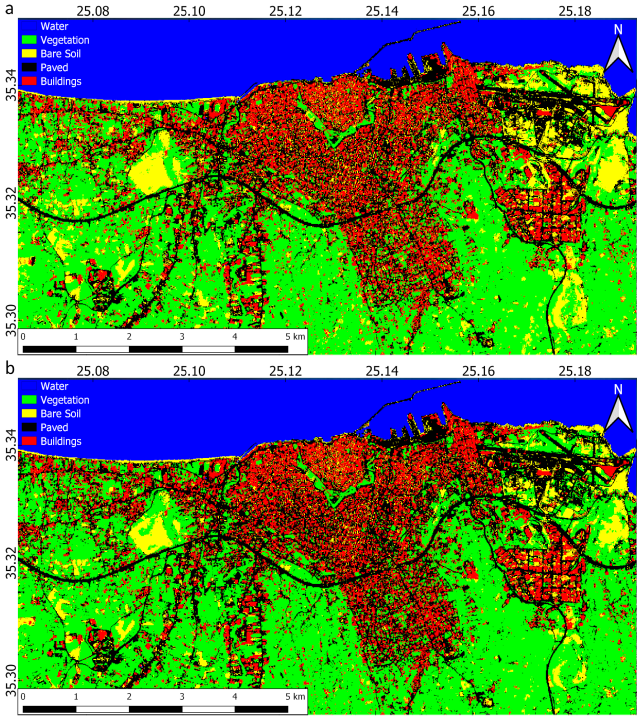


Fig. 9. Land cover maps produced using the RBF kernel for experiment 8B and (a) C-SVC (OA = 84.80) and (b) X-SVC (OA = 86.68).

The information from those atmospheric bands introduce error when included in the training data, resulting in degrading the OA, for both X-SVC and C-SVC. It has to be noted that two of the red edge bands (7 and 8a) do not contribute much to the land cover mapping, in contrast to the SWIR bands (11 and 12) which increase the OA.

Fig. 8 shows the SE versus the OA for all the experiments of Table I for the three kernels. The linear trend lines are shown to allow assumptions for the behavior of OA compared with SE. All three trend lines, corresponding to different Kernels, have positive slopes, and therefore, OA is generally increases when SE increases. The cases of experiment 2B for the polynomial and RBF kernel were not taken into account in the estimation of the trend lines, since these have very low OA, and therefore are considered outliers.

Fig. 9 shows the land cover maps corresponding to the experiments with the highest OA for C-SVC and X-SVC, using the three kernels. The best result in terms of OA is the one produced with X-SVC in experiment 8B using the RBF kernel [Fig. 9(b)] with OA found to be 86.68%. The respective result for C-SVC [Fig. 9(a)] presents an OA of 84.80%. Despite the 2.12% increase in the OA of the resulting land cover map, X-SVC produces a more realistic result, as this appears from visual inspection. C-SVC underestimates the paved surfaces class, while X-SVC succeeds in capturing the road network, even the smaller roads, and the paved surfaces found in the city among buildings, such as parking lots and paved yards. The producer's accuracy (PA) reported for the

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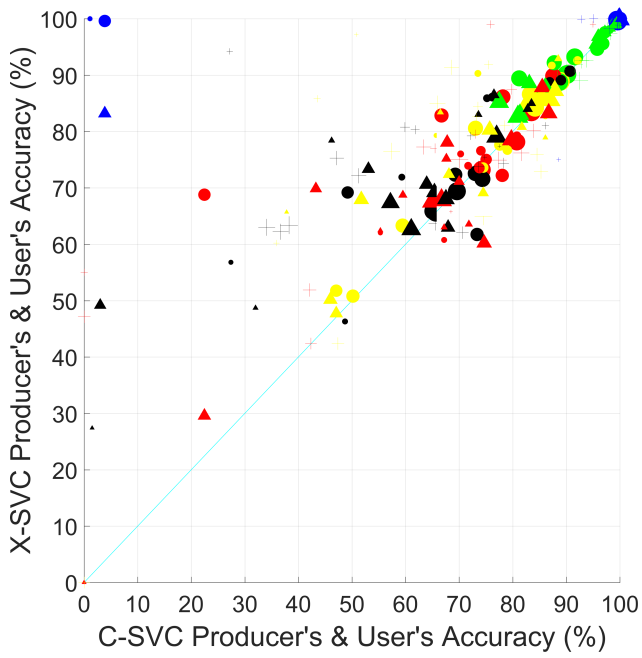


Fig. 10. PA and UA of X-SVC versus C-SVC. Different colors and shapes correspond to different classes and kernels used, respectively: red is for building, black for paved, yellow for bare soil, green for vegetation, blue for water classes, while circle symbol is for RBF, triangle for polynomial, and cross for linear kernels.

paved surfaces is 76.2% and 86.1% for C-SVC and X-SVC, respectively.

Fig. 10 shows the PA and user's accuracy (UA) of X-SVC versus the one of C-SVC for the five land cover types, using three different Kernels and eight different band combinations. Different colors represent different classes, that is, red is for building, black for paved, yellow for bare soil, green for vegetation, and blue for water. Different symbols represent different kernels used, that is, circle corresponds to RBF, triangle to polynomial, and cross to linear. In addition, different symbol sizes indicate the number of classes. X-SVC outperforms C-SVC in most cases (higher PA and UA). Nevertheless, there are cases where the PA or UA of C-SVC is higher than the one of C-SVC. This is mostly true for the experiments involving a small number of bands. The reason is the confusion imposed by the few dimensions, which allows quite reliable results for some classes and consequently, less accurate for the rest. Although for some classes C-SVC presents higher PA or UA than X-SVC (Fig. 10), it has to be noted again that the OA (Fig. 7) of X-SVC is always higher or equal than the one of C-SVC.

V. Conclusion and Outlook

In this study, an extension of the commonly used C-SVC is proposed, which introduces a more effective separation of the training data set, aiming to increase or maintain the accuracy of the classification. We call the new classifier X-SVC, and compared with C-SVC, it does not require a priori determination of the parameter C, which makes its use easier and its application more efficient. X-SVC memorizes the intermediate results of each external iteration during optimization with the SMO algorithm and computes the SE for each case. The optimum solution is then identified based on the optimum SE, which may be found between iterations and not in the final iteration, as assumed by C-SVC.

The performance of the X-SVC in land cover classification was assessed using Sentinel-2 satellite imagery and comparisons were performed against C-SVC. An urban site was selected for testing, to demonstrate the potential of X-SVC in complex environemnts, since urban settings present large diversities in the underlying features and the corresponding spectral responces. Both pure and mixed pixels' spectra corresponding to five land cover types were used as the training data to allow hard classification of the land cover types. The SE and OA were estimated using X-SVC and C-SVC for different combinations of the Sentinel-2 bands and using different kernels. X-SVC outperformed C-SVC or resulted in the same SE and OA in all cases. The largest improvement in terms of OA of X-SVC compared with C-SVC was found to be 0.293.

The experiments performed with real satellite data demonstrated that although the SE of the training data is improving by including more information in terms of spectral bands, this has no impact in OA in some cases.

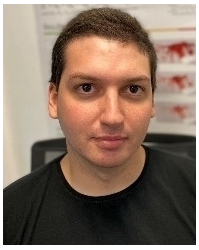
Since SVMs estimate the SVs, which are the "boundary vectors" separating the training data, they rarely manage to completely separate the training data (SE = 1), even with the use of different kernels. This imposes the use of spectral unmixing techniques such as SVR [24] and e-SVM [25] which allow more loose SVs that account for mixed pixels in the training data. These soft classifiers can be proven more effective for land cover classification of urban settings using high-resolution satellite data and the resulting surface cover fraction maps may be more appropriate for other studies, like the ones of urban climate. Therefore, future research will include the extension of X-SVM for spectral unmixing.

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